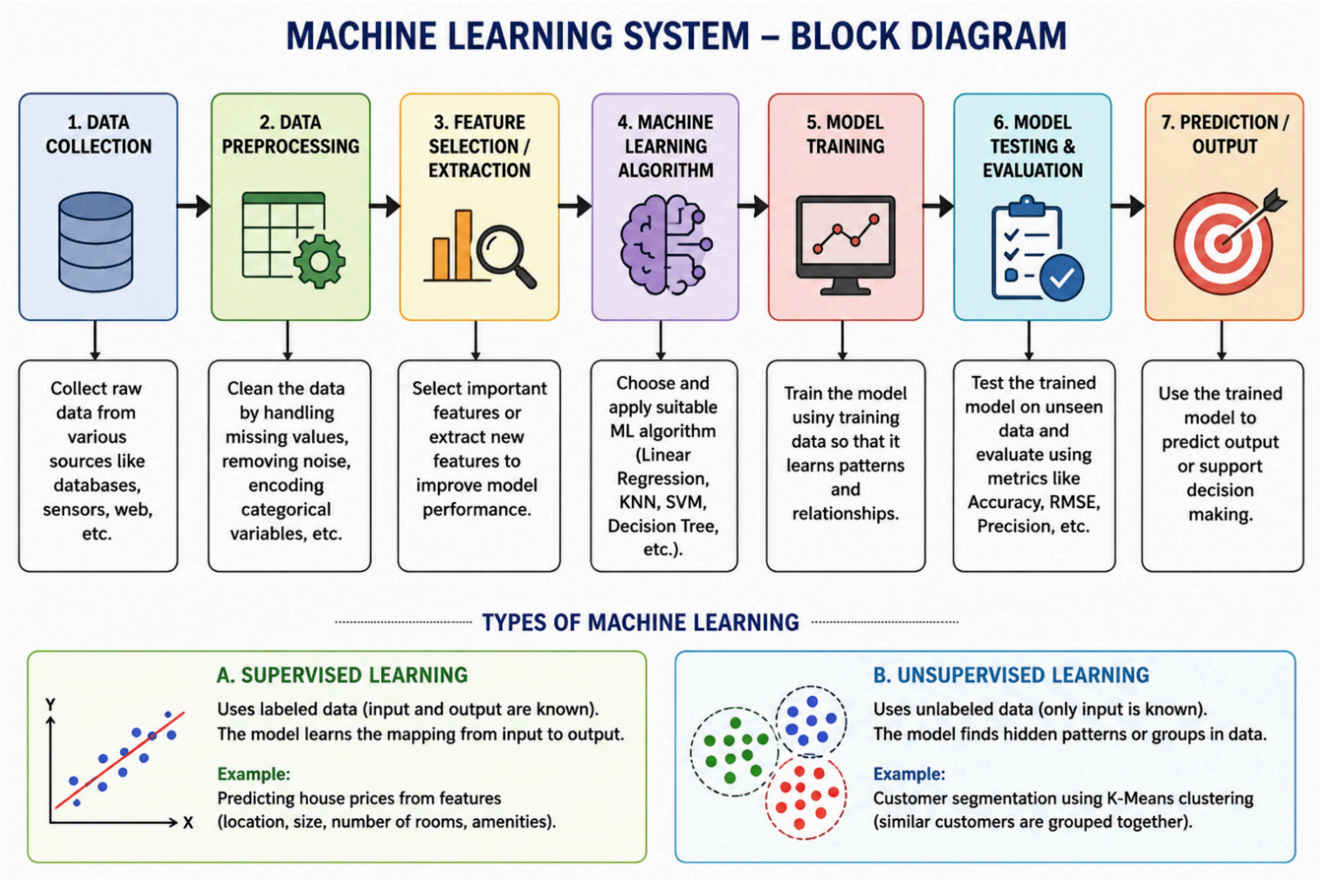
1.With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.



2. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities.
(a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).
(b) Explain feature scaling and handling missing data.
(c) Describe how overfitting can be detected and avoided.

(a) Justify the Choice of Regression Model

Random Forest Regression is the preferred model because:

- Handles complex and non-linear relationships.
- Works well with multiple features such as location, size, rooms, and amenities.
- Produces higher prediction accuracy than Linear Regression.
- Less prone to overfitting than a single Decision Tree.

(Linear Regression is suitable for simple linear relationships, while Decision Trees may overfit.)

(b) Feature Scaling and Handling Missing Data

Feature Scaling

- Converts features to a similar scale using **Normalization** or **Standardization**.
- Improves the performance of algorithms that depend on feature magnitudes.

Handling Missing Data

- Replace missing values with the **mean**, **median**, or **mode**.
- Remove records with too many missing values.
- Use interpolation or prediction methods when appropriate.

(c) Detecting and Avoiding Overfitting

Detection

- Training accuracy is very high, but testing accuracy is low.
- Validation loss increases while training loss decreases.

Avoidance

- Use **cross-validation**.
- Apply **Regularization (L1/L2)**.
- Use **Pruning** for Decision Trees.
- Apply **Early Stopping**.
- Increase the amount of training data.

3. Explain the concept of clustering in unsupervised learning.
(a) Describe the working of the K-Means algorithm with a step-by-step example.
(b) Explain how the value of *K* is chosen using the Elbow Method.
(c) Compare K-Means with Hierarchical Clustering.

Clustering

Clustering is an **unsupervised learning** technique that groups similar data points into clusters without using labeled data.

Applications

- Customer segmentation
- Image segmentation
- Document grouping
- Disease pattern analysis

(a) Working of K-Means Algorithm

Steps

1. Choose the number of clusters (**K**).
2. Randomly initialize **K centroids**.
3. Assign each data point to the nearest centroid.
4. Calculate new centroids by finding the mean of each cluster.

5. Repeat steps 3 and 4 until centroids no longer change.

Example

Suppose there are customer spending values:

10, 12, 15, 70, 75, 80

Choose **K = 2**

- Initial centroids: **10** and **75**
- Cluster 1 → {10, 12, 15}
- Cluster 2 → {70, 75, 80}
- New centroids become **12.3** and **75**
- Repeat until clusters remain unchanged.

Final clusters:

- Cluster 1 → Low spenders
- Cluster 2 → High spenders

(b) Elbow Method

The **Elbow Method** helps determine the optimal value of **K**.

Procedure

1. Run K-Means for different values of **K**.
2. Compute the **Within-Cluster Sum of Squares (WCSS)**.
3. Plot **K** versus **WCSS**.
4. Select the value of **K** where the graph forms an **"elbow"**, indicating diminishing improvement.

(c) K-Means vs Hierarchical Clustering

K-Means	**Hierarchical Clustering**
Requires the value of **K** beforehand	Does not require K initially
Fast for large datasets	Slower for large datasets
Produces flat clusters	Produces a tree-like dendrogram
Easy to implement	Better for understanding hierarchical relationships
Suitable for large datasets	Suitable for small and medium datasets